FACULTY OF INFORMATION TECHNOLOGY
2025 BID DATA AND IOT 600 ASSIGNMENT

RICHFIELD
richfield.ac.za

Name and Surname: Tygran Wozniak

Student ITS No: 402313051

Qualification: Bsc IT 2nd **Year of Study:** 2 **Semester:** 2

Assignment due date: 01-09-2025 **Date submitted:** 01-09-2025

QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

Please tick each box to confirm completion.(10 Marks)

DECLARATION OF ORIGINALITY:
I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

Student Signature: TVW **Date:** 01-09-2025

TABLE OF CONTENTS:

Contents

QUESTION 1 (20 MARKS) 3
QUESTION 2 (20 MARKS) 5
QUESTION 3 (20 MARKS) 6
REFERENCES 7

QUESTION 1 (20 MARKS)

1.1)

Big Data refers to extremely large and complex datasets that exceed the capabilities of traditional IT systems to store, process, and analyse efficiently. It involves high volume, rapid velocity, and diverse variety of data — often requiring distributed computing, parallel processing, and specialized tools like Hadoop, Spark, or NoSQL databases (TechTarget, 2025; Google Cloud, n.d.).

Volume (Size of Data)

Big Data is generated in terabytes to petabytes from sources like sensors, social media, and transaction logs. Traditional tools are not suitable for this scale, as they were designed for smaller, structured datasets (Acropolium, 2024).

Velocity (Speed of Flow of Data)

Data streams in real-time or near real-time, such as stock markets or location data. Most legacy systems cannot handle the speed and frequency of updates required (IBM, n.d.).

Variety (Diverse Data Types and Formats)

Big Data includes structured (SQL tables), semi-structured (JSON, XML), and unstructured (images, video, text) formats. Traditional systems require rigid schemas and struggle with non-tabular formats (TIBCO, n.d.).

Veracity (Uncertainty & Noise)

Data may be incomplete, corrupted, or noisy. Conventional tools expect clean, formatted inputs and cannot handle ambiguity well (Oxford Training Centre, n.d.).

Value (Deriving Meaning from Data)

Big Data enables meaningful insights through advanced analytics and visualization, but traditional tools are limited in scale and flexibility (VinBrain, n.d.).

1.2)

1. Government Databases

Examples include census results, tax records, vehicle registrations, crime reports, and land ownership data.

Data Types: Structured (e.g., ID numbers), semi-structured (e.g., XML forms), and unstructured (e.g., scanned documents).

Uses: Policy planning, fraud detection, and urban development (Acropolium, 2024).

2. Social Media Platforms

Data includes posts, likes, shares, hashtags, and multimedia uploads.

Data Types: Unstructured (text, images), semi-structured (JSON metadata).

Uses: Sentiment analysis, trend forecasting, and behavioral profiling (VinBrain, n.d.).

3. Healthcare Systems

Sources include EHRs, lab results, prescriptions, imaging scans, and wearable data.

Data Types: Structured (diagnosis codes), semi-structured (HL7/FHIR), and unstructured (doctor's notes, radiology images) (TIBCO, n.d.).

Uses: Predictive analytics, personalized medicine, and operational efficiency (VinBrain, n.d.; Oxford Training Centre, n.d.).

QUESTION 2 **(20 MARKS)**

1. Challenges Hospitals Face with Big Data

Hospitals face integration challenges due to diverse formats like structured records and unstructured feedback. Real-time data from IoT devices adds pressure for low-latency processing, which traditional databases cannot handle (IBM, n.d.). Data quality issues such as missing or duplicate entries can distort diagnoses. Privacy laws like POPIA and HIPAA require strict controls, and breaches can result in legal consequences (Oxford Training Centre, n.d.).

2. Common Tools Used in the Field

Tools/Program	Use Case
Apache Hadoop	Distributed processing
Google BigQuery	Scalable cloud data and warehousing
NVIDIA Clara	Medical imaging & AI diagnostics
Power BI	Data visualisation
Azure IoT Hub	IoT integration

3. How Should Hospitals Clean and Organize Their Data?

Hospitals should standardize formats using HL7 and FHIR to ensure interoperability (TIBCO, n.d.). Deduplication tools should be used to prevent statistical distortion. Missing data must be flagged or imputed to preserve integrity. Metadata tagging helps organize unstructured files like MRI scans. IoT monitoring platforms such as Azure IoT Hub ensure device reliability and data accessibility (IBM, n.d.).

4. How Should Hospitals Keep Their Data Secure?

Security requires layered strategies aligned with HIPAA and POPIA. Role-based access control ensures only authorized personnel can view sensitive data. The 21st Century Cures Act mandates secure disclosures, and analytics tools must operate within compliant environments (Oxford Training Centre, n.d.).

5. How Can Hospitals Use This Data to Understand Past Trends and Predict Future Needs?

Hospitals can use Apache Spark, SQL, and Python (Pandas) to analyze historical data. Visualization tools like Power BI and Tableau help identify trends and optimize resource allocation (HashStudioz, 2025). Predictive analytics using machine learning can forecast events like patient readmissions, improving care delivery (VinBrain, n.d.).

QUESTION 3 **(20 MARKS)**

An IoT system begins with sensors and devices that collect real-world data such as temperature, speed, or location. These inputs are digitized and fed into Big Data platforms for analysis (IBM, n.d.). The connectivity layer enables transmission between devices and processing systems using protocols like Wi-Fi, Bluetooth, or LPWAN. Reliable connectivity is essential for real-time applications, as poor transmission can lead to data loss or latency (IBM, n.d.).

Once transmitted, data enters the processing infrastructure, which may be cloud-based or local. Here, it is stored, formatted, and analyzed using platforms like Apache Spark, AWS IoT Core, or Azure Stream Analytics (TechTarget, 2025). This layer is responsible for converting raw data into actionable insights, detecting anomalies, and enabling automation.

Finally, the processed data is visualized through user interfaces such as dashboards or mobile apps. These interfaces allow users to interact with devices, monitor performance, and make informed decisions. Together, these components form a continuous data flow — from collection to transmission, processing, and action — each playing a vital role in the effectiveness of an IoT system (IBM, n.d.).

REFERENCES:

- Certainly, Tygran — here are the references formatted in Harvard style using today's date (1 September 2025):
- 📚 Harvard References
- Acropolium (2024) 9 Big Data Use Cases Across Major Industries [2025 Guide]. Available at: https://acropolium.com/blog/big-data-use-cases-across-major-industries/ (Accessed: 1 September 2025).
- Google Cloud (n.d.) What is Big Data?. Available at: https://cloud.google.com/learn/what-is-big-data (Accessed: 1 September 2025).
- HashStudioz Technologies (2025) Healthcare Dashboards That Matter: Leveraging Tableau & Power BI for Patient Insights. Available at: https://www.hashstudioz.com/blog/healthcare-dashboards-that-matter-leveraging-tableau-power-bi-for-patient-insights/ (Accessed: 1 September 2025).
- IBM (2024) What is Hadoop Distributed File System (HDFS)?. Available at: https://www.ibm.com/think/topics/hdfs (Accessed: 1 September 2025).
- IBM (n.d.) IoT Data Flow and Lifecycle – Maximo Asset Monitor. Available at: https://www.ibm.com/docs/en/maximo-asset-monitor?topic=features-iot-data-flow-lifecycle (Accessed: 1 September 2025).
- Oxford Training Centre (n.d.) Challenges of Big Data in Healthcare and Their Impact on Patient Care. Available at: https://oxfordcentre.uk/resources/healthcare-management/challenges-of-big-data-in-healthcare-and-their-impact-on-patient-care/ (Accessed: 1 September 2025).
- TechTarget (2025) 18 Top Big Data Tools and Technologies to Know About in 2025. Available at: https://www.techtarget.com/searchdatamanagement/feature/15-big-data-tools-and-technologies-to-know-about (Accessed: 1 September 2025).
- TIBCO (n.d.) What is HL7 FHIR?. Available at: https://www.tibco.com/glossary/what-is-hl7-fhir (Accessed: 1 September 2025).
- VinBrain (n.d.) Top 10 Big Data Applications in Healthcare – Part 1. Available at: https://vinbrain.net/top-10-big-data-application-in-healthcare-part-1 (Accessed: 1 September 2025).